LIST

In [1]:
```
l = []
l
```

Out[1]:
```
[]
```

In [2]:
```
type(l)
```

Out[2]:
```
list
```

In [3]:
```
len(l)
```

Out[3]:
```
0
```

In [4]:
```
l.append(10)
```

In [5]:
```
l
```

Out[5]:
```
[10]
```

In [6]:
```
l.append(20)
l.append(30)
l.append(40)
l.append(50)
```

In [7]:
```
l
```

Out[7]:
```
[10, 20, 30, 40, 50]
```

In [8]:
```
len(l)
```

Out[8]:
```
5
```

In [9]:
```
l.append(10)
l
```

Out[9]:
```
[10, 20, 30, 40, 50, 10]
```

In [10]:
```
l1 = []
l1
```

Out[10]:
```
[]
```

In [11]:
```
l1 = l.copy()
l1
```

Out[11]:
```
[10, 20, 30, 40, 50, 10]
```

In [12]:
```
l == l1
```

Out[12]:
```
True
```

In [13]:
```
print(id(l))
print(id(l1))
```

```
1813542346304
1813567461760
```

In [14]:
```
print(l)
print(l1)
```

```
[10, 20, 30, 40, 50, 10]
[10, 20, 30, 40, 50, 10]
```

In [15]:
```
l2 = []
l2
```

Out[15]:
```
[]
```

In [16]:
```
l2.append(2)
l2.append(2.3)
l2.append(True)
l2.append(1+2j)
l2.append('nit')
l2.append([1,2,3])
```

In [17]:
```
l2 #nested list
```

Out[17]:
```
[2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]
```

In [18]:
```
print(l)
print(l1)
print(l2)
```

```
[10, 20, 30, 40, 50, 10]
[10, 20, 30, 40, 50, 10]
[2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]
```

In [19]:
```
l
```

Out[19]:
```
[10, 20, 30, 40, 50, 10]
```

In [20]:
```
l.clear()
```

In [21]:
```
l
```

Out[21]:
```
[]
```

In [22]:
```
del l
```

In [23]:
```
l
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[23], line 1
----> 1 l

NameError: name 'l' is not defined
```

In [24]:
```
print(l1)
```

```
print(l2)
```

```
[10, 20, 30, 40, 50, 10]
[2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]
```

In [25]:
```
l1
```

Out[25]:
```
[10, 20, 30, 40, 50, 10]
```

In [26]:
```
l1.count(20)
```

Out[26]:
```
1
```

In [27]:
```
l1.count(10)
```

Out[27]:
```
2
```

In [28]:
```
print(l1)
print(l2)
```

```
[10, 20, 30, 40, 50, 10]
[2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]
```

In [29]:
```
l1.extend(l2)
```

In [30]:
```
print(l1)
print(l2)
```

```
[10, 20, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]
[2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]
```

In [31]:
```
l1.index(200)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[31], line 1
----> 1 l1.index(200)

ValueError: 200 is not in list
```

In [32]:
```
l2.index([1,2,3])
```

Out[32]:
```
5
```

In [33]:
```
l2
```

Out[33]:
```
[2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]
```

In [34]:
```
for i in l2:
    print(i)
```

```
2
2.3
True
(1+2j)
nit
[1, 2, 3]
```

In [35]:
```
l1
```

Out[35]: [10, 20, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]

In [36]:
```
l1.insert(1, 15)
```

In [37]:
```
l1
```

Out[37]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]

In [38]:
```
sal = [5000, 5000, 10000, ]
sal
```

Out[38]: [5000, 5000, 10000]

In [39]:
```
sal[2]
```

Out[39]: 10000

In [40]:
```
l1
```

Out[40]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]

In [41]:
```
l1.insert(3, 23)
l1
```

Out[41]: [10, 15, 20, 23, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]

In [42]:
```
l1
```

Out[42]: [10, 15, 20, 23, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit', [1, 2, 3]]

In [43]:
```
l1.pop()
```

Out[43]: [1, 2, 3]

In [44]:
```
l1
```

Out[44]: [10, 15, 20, 23, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit']

In [45]:
```
l1.pop(3)
```

Out[45]: 23

In [46]:
```
l1
```

Out[46]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit']

In [47]:
```
l1.pop(20)
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[47], line 1
----> 1 l1.pop(20)

IndexError: pop index out of range
```

In [48]:
```
l1
```

Out[48]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, (1+2j), 'nit']

In [49]:
```
l1.remove(200)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[49], line 1
----> 1 l1.remove(200)

ValueError: list.remove(x): x not in list
```

In [50]:
```
l1.remove((1+2j))
```

In [52]:
```
l1
```

Out[52]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [53]:
```
l1[:]
```

Out[53]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [54]:
```
len(l1)
```

Out[54]: 11

In [55]:
```
l1
```

Out[55]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [56]:
```
l1[2:]
```

Out[56]: [20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [57]:
```
l1
```

Out[57]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [58]:
```
l1[:2]
```

Out[58]: [10, 15]

In [59]:
```
l1
```

Out[59]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [60]:
```
l1[::2]
```

Out[60]: [10, 20, 40, 10, 2.3, 'nit']

In [61]:
```
l1[::3]
```

Out[61]: [10, 30, 10, True]

In [62]:
```
l1
```

Out[62]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [63]:
```
l1[2:10:4]
```

Out[63]:
```
[20, 10]
```

In [64]:
```
l1
```

Out[64]:
```
[10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']
```

In [65]:
```
l1[::-1]
```

Out[65]:
```
['nit', True, 2.3, 2, 10, 50, 40, 30, 20, 15, 10]
```

In [66]:
```
l1
```

Out[66]:
```
[10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']
```

In [67]:
```
l1[::-2]
```

Out[67]:
```
['nit', 2.3, 10, 40, 20, 10]
```

In [68]:
```
l1
```

Out[68]:
```
[10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']
```

In [69]:
```
l1[::-5]
```

Out[69]:
```
['nit', 50, 10]
```

In [70]:
```
l1
```

Out[70]:
```
[10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']
```

In [76]:
```
l2 = ['a', 'b','c']
```

In [77]:
```
l2
```

Out[77]:
```
['a', 'b', 'c']
```

In [78]:
```
l2[-4]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[78], line 1
----> 1 l2[-4]

IndexError: list index out of range
```

In [80]:
```
l2[-4:]
```

Out[80]:
```
['a', 'b', 'c']
```

In [81]:
```
l2
```

Out[81]:
```
['a', 'b', 'c']
```

In [82]:
```
l2[0]
```

Out[82]: 'a'

In [83]:
```
l2[0] = 'nit'
```

In [84]:
```
l2
```

Out[84]: ['nit', 'b', 'c']

In [85]:
```
l1
```

Out[85]: [10, 15, 20, 30, 40, 50, 10, 2, 2.3, True, 'nit']

In [86]:
```
l1[4] = 400
```

In [87]:
```
l1
```

Out[87]: [10, 15, 20, 30, 400, 50, 10, 2, 2.3, True, 'nit']

In [88]:
```
l1
```

Out[88]: [10, 15, 20, 30, 400, 50, 10, 2, 2.3, True, 'nit']

In [89]:
```
l1.index('nit')
```

Out[89]: 10

In [90]:
```
l1
```

Out[90]: [10, 15, 20, 30, 400, 50, 10, 2, 2.3, True, 'nit']

In [91]:
```
l1[10]
```

Out[91]: 'nit'

In [93]:
```
# nested indexing ( index inside index)

print(l1[10][0])
print(l1[10][1])
print(l1[10][2])
```

```
n
i
t
```

In [94]:
```
l1
```

Out[94]: [10, 15, 20, 30, 400, 50, 10, 2, 2.3, True, 'nit']

In [95]:
```
l1.reverse()
```

In [96]:
```
l1
```

Out[96]: ['nit', True, 2.3, 2, 10, 50, 400, 30, 20, 15, 10]

In [97]:
```
l1
```

Out[97]: ['nit', True, 2.3, 2, 10, 50, 400, 30, 20, 15, 10]

In [98]:
```
l2
```

Out[98]: ['nit', 'b', 'c']

In [99]:
```
l2.sort()
```

In [100…
```
l2
```

Out[100… ['b', 'c', 'nit']

In [101…
```
l2.sort(reverse=True)
l2
```

Out[101… ['nit', 'c', 'b']

In [102…
```
l1
```

Out[102… ['nit', True, 2.3, 2, 10, 50, 400, 30, 20, 15, 10]

In [103…
```
l1.sort()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[103], line 1
----> 1 l1.sort()

TypeError: '<' not supported between instances of 'bool' and 'str'
```

In [104…
```
l1
```

Out[104… ['nit', True, 2.3, 2, 10, 50, 400, 30, 20, 15, 10]

In [105…
```
l2
```

Out[105… ['nit', 'c', 'b']

In [106…
```
for i in l2:
    print(i)
```

```
nit
c
b
```

In [111…
```
for i in enumerate(l2): # paired items
    print(i)
```

```
(0, 'nit')
(1, 'c')
(2, 'b')
```

List is completed

TUPLE

In [112…
```
t = ()
t
```

Out[112… ()

In [1]:
```
t1 = (10, 20, 30, 40, 50, 50)
t1
```

Out[1]: (10, 20, 30, 40, 50, 50)

In [2]:
```
t2 = (10, 'prakash', 1+2j, True, 2.3)
t2
```

Out[2]: (10, 'prakash', (1+2j), True, 2.3)

In [3]:
```
print(t1)
print(t2)
```

```
(10, 20, 30, 40, 50, 50)
(10, 'prakash', (1+2j), True, 2.3)
```

In [5]:
```
print(type(l1))
print(type(t1))
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[5], line 1
----> 1 print(type(l1))
      2 print(type(t1))

NameError: name 'l1' is not defined
```

In [6]:
```
t1
```

Out[6]: (10, 20, 30, 40, 50, 50)

In [7]:
```
t1.count(50)
```

Out[7]: 2

In [8]:
```
t1.index(10)
```

Out[8]: 0

In [9]:
```
t1
```

Out[9]: (10, 20, 30, 40, 50, 50)

In [10]:
```
t1[:4]
```

Out[10]: (10, 20, 30, 40)

In [11]:
```
t1
```

Out[11]: (10, 20, 30, 40, 50, 50)

In [12]:
```
t1[5]
```

Out[12]:
```
50
```

In [124…
```
t1[5] = 60
t1
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[124], line 1
----> 1 t1[5] = 60
      2 t1

TypeError: 'tuple' object does not support item assignment
```

In [125…
```
icici = (123, 'cizps43443')
icici
```

Out[125…
```
(123, 'cizps43443')
```

In [126…
```
icici[0] = 456
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[126], line 1
----> 1 icici[0] = 456

TypeError: 'tuple' object does not support item assignment
```

In [127…
```
t1
```

Out[127…
```
(10, 20, 30, 40, 50, 50)
```

In [128…
```
t1 * 3
```

Out[128…
```
(10, 20, 30, 40, 50, 50, 10, 20, 30, 40, 50, 50, 10, 20, 30, 40, 50, 50)
```

TUPLE WE ARE COMPLETED

3rd mar -- SET

In [13]:
```
s = {}
s
```

Out[13]:
```
{}
```

In [14]:
```
type(s)
```

Out[14]:
```
dict
```

In [15]:
```
s = set()
s
```

Out[15]: set()

In [16]:
```
type(s)
```

Out[16]: set

In [18]:
```
s1 = {100, 25, 50, 5, 20, 30}
s1
```

Out[18]: {5, 20, 25, 30, 50, 100}

In [19]:
```
type(s1)
```

Out[19]: set

In [20]:
```
len(s1)
```

Out[20]: 6

In [21]:
```
s1
```

Out[21]: {5, 20, 25, 30, 50, 100}

In [22]:
```
s1.add(15, 35)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[22], line 1
----> 1 s1.add(15, 35)

TypeError: set.add() takes exactly one argument (2 given)
```

In [26]:
```
s1.add(15)
```

In [35]:
```
s1
```

Out[35]: {5, 15, 20, 25, 30, 50, 100}

In [29]:
```
s2 = s1.copy()
```

In [34]:
```
s2
```

Out[34]: {5, 10, 15, 20, 25, 30, 50, 100}

In [32]:
```
print(id(s1))
print(id(s2))
```

```
2083523038112
2083523039008
```

In [33]:
```
s2.add(10)
```

In [36]:
```
s2[0]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[36], line 1
----> 1 s2[0]

TypeError: 'set' object is not subscriptable
```

In [37]:
```
s2
```

Out[37]:
```
{5, 10, 15, 20, 25, 30, 50, 100}
```

In [39]:
```
s2[:]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[39], line 1
----> 1 s2[:]

TypeError: 'set' object is not subscriptable
```

In [40]:
```
s3 = {'a', 1+2j, 2.3, True}
s3
```

Out[40]:
```
{(1+2j), 2.3, True, 'a'}
```

In [41]:
```
print(s1)
print(s2)
print(s3)
```

```
{50, 100, 5, 20, 25, 30, 15}
{50, 100, 5, 20, 25, 10, 30, 15}
{True, 2.3, 'a', (1+2j)}
```

In [44]:
```
s1
```

Out[44]:
```
{5, 15, 20, 25, 30, 100}
```

In [45]:
```
s1.pop()
```

Out[45]:
```
100
```

In [46]:
```
s1.pop(3)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[46], line 1
----> 1 s1.pop(3)

TypeError: set.pop() takes no arguments (1 given)
```

In [50]:
```
s1
```

Out[50]:
```
{15, 20, 25, 30}
```

In [49]:
```
s1.discard(5)
```

In [51]:
```
s1.remove(150)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[51], line 1
----> 1 s1.remove(150)

KeyError: 150
```

In [56]:
```
s1
```

Out[56]:
```
{15, 20, 25, 30}
```

In [53]:
```
s1.discard(100)
```

In [54]:
```
s1
```

Out[54]:
```
{15, 20, 25, 30}
```

In [55]:
```
s1.add(30)
s1
```

Out[55]:
```
{15, 20, 25, 30}
```

SET OPERATION

In [58]:
```
a1 = {1,2,3,4,5}
b1 = {4,5,6,7,8}
c1 = {8,9,10}
```

In [59]:
```
a1.union(b1)
```

Out[59]:
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [60]:
```
a1.union(b1, c1)
```

Out[60]:
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

In [61]:
```
a1 | b1
```

Out[61]:
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [62]:
```
b1 | c1
```

Out[62]:
```
{4, 5, 6, 7, 8, 9, 10}
```

In [63]:
```
a1 | b1 | c1
```

Out[63]:
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

intersection

In [64]:
```
print(a1)
print(b1)
```

```
print(c1)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [66]:
```
a1.intersection(b1)
```

Out[66]:
```
{4, 5}
```

In [67]:
```
a1.intersection(c1)
```

Out[67]:
```
set()
```

In [68]:
```
a1 & b1
```

Out[68]:
```
{4, 5}
```

In [69]:
```
b1 & c1
```

Out[69]:
```
{8}
```

In [70]:
```
a1 & c1
```

Out[70]:
```
set()
```

In [71]:
```
a1 = {1,2,3,4,5}
b1 = {4,5,6,7,8}
c1 = {8,9,10}
```

In [72]:
```
a1.difference(b1)
```

Out[72]:
```
{1, 2, 3}
```

In [74]:
```
b1 - c1
```

Out[74]:
```
{4, 5, 6, 7}
```

In [75]:
```
print(a1)
print(b1)
print(c1)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [76]:
```
a1.difference(b1)
```

Out[76]:
```
{1, 2, 3}
```

In [77]:
```
a1.symmetric_difference(b1)  # remove common element and show rest element from
```

Out[77]:
```
{1, 2, 3, 6, 7, 8}
```

In [78]:
```
a1 ^ b1
```

Out[78]:
```
{1, 2, 3, 6, 7, 8}
```

In [79]:
```
print(a1)
print(b1)
print(c1)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [80]:
```
a1.difference(b1)
```

Out[80]:
```
{1, 2, 3}
```

In [81]:
```
a1
```

Out[81]:
```
{1, 2, 3, 4, 5}
```

In [83]:
```
a1.difference_update(b1)
```

In [84]:
```
a1
```

Out[84]:
```
{1, 2, 3}
```

In [85]:
```
print(a1)
print(b1)
print(c1)
```

```
{1, 2, 3}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

superset | subset | disjoint

In [82]:
```
s5 = {1,2,3,4,5,6,7,8,9}
s6 = {3,4,5,6,7,8}
s7 = {10,20,30,40}
```

In [86]:
```
s5.issuperset(s6)
```

Out[86]:
```
True
```

In [89]:
```
s6.issuperset(s5)
```

Out[89]:
```
False
```

In [90]:
```
s5 = {1,2,3,4,5,6,7,8,9}
s6 = {3,4,5,6,7,8}
s7 = {10,20,30,40}
```

In [88]:
```
s6.issubset(s5)
```

Out[88]:
```
True
```

In [69]:
```
s7.isdisjoint(s5)
```

Out[69]:
```
True
```

In [87]:
```
s7.isdisjoint(s6)
```

Out[87]: True

In [71]:
```
s5 = {1,2,3,4,5,6,7,8,9}
s6 = {3,4,5,6,7,8}
s7 = {10,20,30,40}
```

In [72]:
```
s5.isdisjoint(s6)
```

Out[72]: False

In [75]:
```
a3 = {1,2,3,4,5,6,7,8,9}
b3 = {13,14,15,16,17,18}
c3 = {10,20,30,40}
```

In [79]:
```
b3.issubset(a3)
```

Out[79]: False

In [81]:
```
b3.isdisjoint(c3)
```

Out[81]: True

set we are completed

RANGE

In [ ]:
```

```